

Appendix

Anonymous submission

A Patient System

Listing 1: Keyword Mapping Dictionary (example)

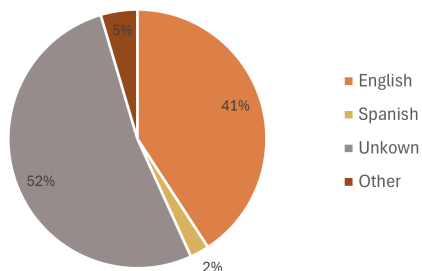
```
keyword_mapping = {
    # Demographics
    ('age',): ('Demographics', 'age'),
    ('language', 'english', 'spanish'): ('Demographics', 'language'),
    ('religion', 'religious'): ('Demographics', 'religion'),
    ('marital', 'married', 'single', 'divorced', 'widowed'): ('Demographics', 'marital_status'),
    ('gender', 'sex'): ('Demographics', 'gender'),
    ('insurance',): ('Demographics', 'insurance'),
    ('ethnicity',): ('Demographics', 'ethnicity'),
    # Medications
    ('prescription', 'medications', 'medication', 'drugs'): ('Prescription', None),
    ('admission medications', 'initial meds'): ('Medications on Admission', None),
    # Procedures
    ('procedure', 'surgery', 'operation'): [('Procedure', None), ('Major Surgical or Invasive Procedure', None)],
    # Imaging and reports
    ('electrocardiogram', 'ecg'): ('ECG', None),
    ('echocardiogram', 'echo'): ('Echo', None),
    ('radiology', 'x-ray', 'ct', 'mri', 'imaging'): ('Radiology', None),
    # History
    ('hpi', 'present illness', 'history of present illness'): ('History of Present Illness', None),
    ('past medical', 'pmh', 'past medical history'): ('Past Medical History', None),
    ('family history',): ('Family History', None),
    ('social history', 'drinking', 'smoking', 'drug use', 'tobacco', 'alcohol'): ('Social History', None),
    # Allergies
    ('allergy', 'allergies', 'allergic'): ('Allergies', None),
    # Physical exam
    ('heent', 'head', 'eyes', 'ears', 'nose', 'throat'): ('Physical Exam.Admission', 'HEENT'),
    ('physical exam',): ('Physical Exam.Admission', None),
    ...}
```

Listing 2: Patient JSON example (**** indicates placeholders for privacy issues.)

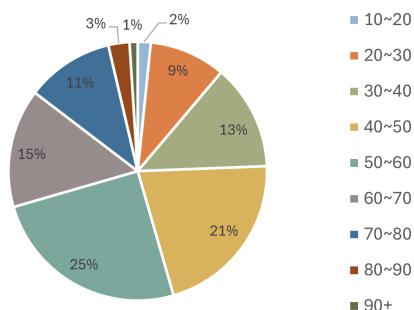
```
"Patient": {
  "Admission_info": {
    "patient_id": ****,
    "admission_id": ****,
    "admission_diagnosis": "arrhythmia"
  },
  "Demographics": {
    "insurance": "private",
    "language": "engl",
    "marital_status": "****",
    "ethnicity": "white",
    "gender": "****",
    "age": ****
  },
  "Diagnoses": [
    "4019",
    "Hypertension NOS",
    "Unspecified essential hypertension"
  ],
  "Prescription": [
    "Sodium Chloride 0.9% Flush",
    "Lisinopril",
    "Heparin",
    ...
  ],
  "Introduction": "Hi, I'm a ****. I was referred here by my clinic ...my doctor was concerned about a possible arrhythmia.",
  "ECG": [
    "2105-03-03",
    "Atrial pacing and A-V conduction which is new compared to previous tracings."
  ],
  "Radiology": [
    {
      "time": "2105-03-03",
      "part": "CHEST (PA & LAT)",
      "medical condition": "**** with new dual chamber",
      "ppm reason for this examination": "Evaluate lead position",
      "final report history": "Pacemaker placement.",
      "findings": "In comparison with the study of ****, there has been placement of ... other acute cardiopulmonary disease."
    }
  ],
  "Allergies": "No Known Allergies / Adverse Drug Reactions",
  "Chief Complaint": "Fatigue, lightheadedness, bradycardia, sinus pauses",
  "Major Surgical or Invasive Procedure": "Pacemaker placement (St. Medical Accent PM2210 dual chamber pacemaker)",
  "Physical Exam": {
    "Admission": {
      "VS": "T=98.0 BP=158/91 HR=61 RR=18 O2 sat=95",
      "General": "WDWN M in NAD. Oriented x3. Mood, affect appropriate. Fit",
      "HEENT": "NCAT. Sclera anicteric. PERRL, EOMI. Conjunctiva were pink, no pallor or cyanosis of the oral mucosa. No xanthoema.",
      "Neck": "Supple with JVP below clavicle at 90 degrees.",
      "Cardiac": "PMI located in 5th intercostal space, midclavicular line. RR, normal S1, S2. No m/r/g. No thrills, lifts. No S3 or S4.",
      "Lungs": "No chest wall deformities, scoliosis or kyphosis. Resp were unlabored, no accessory muscle use. CTAB, no crackles, wheezes or rhonchi.",
      "Abdomen": "Soft, NTND. No HSM or tenderness. Abd aorta not enlarged by palpation. No abdominal bruits.",
      "Extremities": "No c/c/e. No femoral bruits.",
      "Skin": "No stasis dermatitis, ulcers, scars, or xanthomas.",
      "Pulses": {
        "Right": "Carotid 2+ Femoral 2+ Popliteal 2+ DP 2+ PT 2+",
        "Left": "Carotid 2+ Femoral 2+ Popliteal 2+ DP 2+ PT 2+"
      }
    },
    ...
  ],
  "Respiratory": {
    "O2 saturation pulseoxymetry": [
      ["2105-02-28 03:15:00", "97.0 %"]
    ],
    ...
  }
}
```

B MIMIC Patient Demographics

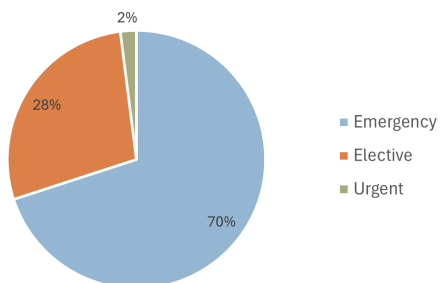
Distribution of Language



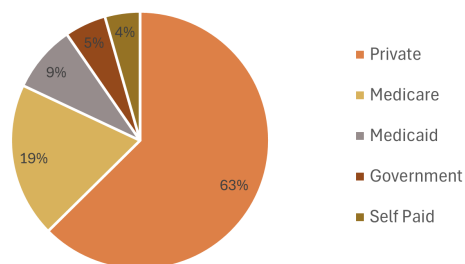
Distribution of Age



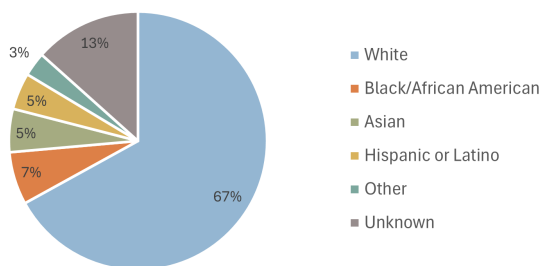
Distribution of Admission Type



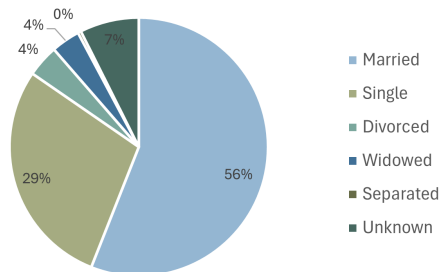
Distribution of Insurance

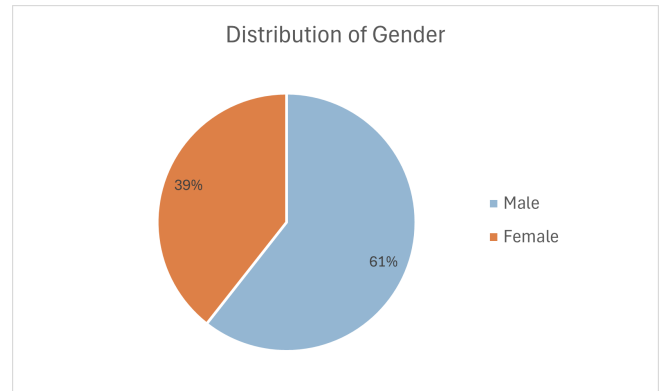
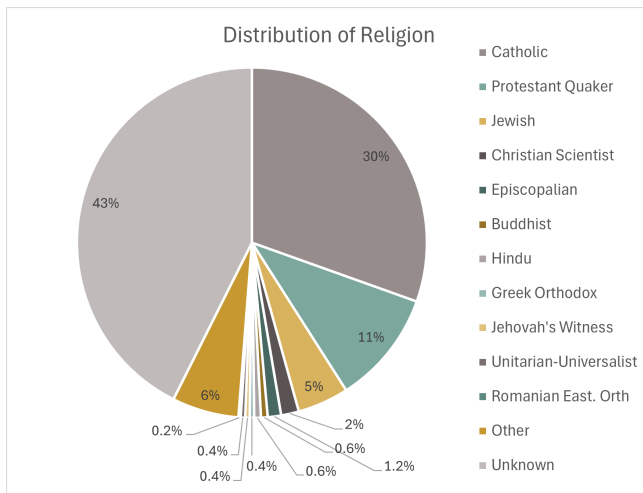


Distribution of Ethnicity



Distribution of Marital Status





C Doctor Agent on MIMIC Open-Ended Diagnosis

Listing 3: Initial Specialist Triage Prompt

You are a general practitioner triaging a new patient. Based on the patient's initial admission information, recommend one or more medical specialists to consult.

The maximum number of specialists is 5.

Return your answer in the following JSON format only:

```
{
  "RATIONALE": "<short justification>",
  "SUGGEST_SPECIALISTS": [<list of specialists>]
}
```

Listing 4: Central Agent Coordination Prompt

You are a central medical coordinator overseeing a diagnostic team.

Based on the current patient case and the specialists already involved, decide if additional experts are needed or if any can be removed.

Only suggest adding new specialists if there's missing domain knowledge. Only suggest removing specialists if their role has been fully covered or no longer needed.

The maximum number of specialists is 5.

Respond in this JSON format only:

```
{
  "ADD": [<specialists to add>],
  "REMOVE": [<specialists to remove>],
  "UPDATED_LIST": [<updated specialists>],
  "RATIONALE": "<short justification>"
}
```

Listing 5: Specialist Confidence Rating Prompt

You are a {spec}. Based on the current patient case, decide whether you are confident enough to make a diagnosis or whether more information is needed.

Choose between the following ratings:

"Very Confident"- The diagnosis is strongly supported by current information, and no major uncertainties remain.

"Somewhat Confident"- The diagnosis is likely given the evidence, but a bit more information would increase certainty.

"Neither Confident or Unconfident"- Some clues suggest a possible diagnosis, but key details are still missing.

"Somewhat Unconfident"- Several diagnoses remain plausible; more data is needed to narrow them down.

"Very Unconfident"- There is too little evidence to form a reasonable diagnostic opinion.

Respond in the following format only:

DECISION: chosen rating from the above list.

Listing 6: Solo Specialist Diagnosis or Question Prompt

You are a {spec}. Based on the current patient case, list the top 10 most likely diagnoses for this patient.

Only include the **diagnosis name** in the list.

Respond in this JSON format only:

```
{
  "RESPONSE_TYPE": "diagnosis",
  "RESPONSE_CONTENT": "[<your diagnosis list>]",
  "RATIONALE": "<brief justification>"
}
```

Listing 7: Solo Specialist Follow-Up Question Prompt

You are a {spec}. Based on your medical expertise and the current information, propose the most important next question that would help you narrow down or confirm a diagnosis.

The question should be specific and relevant to the case.

Do not repeat any questions from the previous conversation log or ask about information already provided.

Avoid asking about topics already answered with "I don't know".

If referencing labs, vitals, ECG, radiology, etc. - which may have multiple time points - be clear about the desired time window.

Respond in this JSON format only:

```
{
  "RESPONSE_TYPE": "question",
  "RESPONSE_CONTENT": "<your follow-up question>",
  "RATIONALE": "<brief justification>"
}
```

Listing 8: Collaborative Decision Proposal Prompt

You are a {spec}. Based on your medical expertise and the current information, respond with either:

- A list of the top 10 most likely diagnoses (if you believe you can make a clinical judgment now), or
- A follow-up question (if you believe more information is needed to proceed)

Only include the ****diagnosis name**** in the list.

The question should be specific and relevant to the case.

Do not repeat any questions from the previous conversation log or ask about information already provided.

Avoid asking about topics already answered with "I don't know".

If referencing labs, vitals, ECG, radiology, etc. - which may have ****multiple time points**** - be clear about the ****desired time window****.

Respond in this JSON format only:

```
{
  "RESPONSE_TYPE": "<diagnosis | question>",
  "RESPONSE_CONTENT": "<your diagnosis list or follow-up question>",
  "CONFIDENCE": "<1-5>", // 5 = Very confident on the appropriateness of the response,
  "RATIONALE": "<brief justification>"
}
```

Listing 9: Voting Prompt

You are a {voter['SPECIALIST']}. Another specialist ({candidate['SPECIALIST']}) proposed the following:

They suggested to proceed with a ****{candidate['RESPONSE_TYPE']}**, which is:

"{candidate['RESPONSE_CONTENT']}"

Their rationale is:

"{candidate['RATIONALE']}"

Based on the current patient case, decide if you agree with the proposed next step.

Respond with ONLY one of the following options:

- AGREE
 - DISAGREE
-

D Walkthrough

This section provides a walkthrough of the full workflow through one specific patient case:

- **Initialize Visit Log**

Patient System generates initial information: *"Hi, I'm here with my newborn daughter. She was born recently and we've been referred here because her bilirubin levels are high."*

- **Specialist Team Formation**

The Central Agent receives the initial visit log. Based on this input, the Central Agent initializes an appropriate Specialist Team tailored to the current case complexity.

Suggest Specialists: *"Pediatrician", "Pediatric Hepatologist"*

Rationale: *"The newborn has elevated bilirubin levels, indicating neonatal jaundice, which may require specialized evaluation to rule out underlying causes and prevent complications such as kernicterus."*

- **Specialist Response Generation – Multi-agent**

- Specialist 1:

Pediatrician: *"Can you provide the newborn's age in hours or days, and the most recent total and direct bilirubin levels?"*

Rationale: *"The age of the newborn and specific bilirubin values are critical for assessing the risk of hyperbilirubinemia and determining the urgency and type of intervention needed."* - Confidence: 5

- Specialist 2:

Pediatric Hepatologist: *"Can you tell me the baby's gestational age at birth, current age in days, and whether she is breastfeeding or formula feeding?"*

Rationale: *"The timing of jaundice onset, feeding method, and gestational age are critical for differentiating between physiologic jaundice, breastfeeding jaundice, and more serious causes such as hemolytic disease or metabolic disorders."* - Confidence: 5

- Specialist Voting:

"Pediatric Hepatologist" agrees with "Pediatrician", process with question by "Pediatrician".

Rationale: *"The age of the newborn and specific bilirubin values are critical for assessing the risk of hyperbilirubinemia and determining the urgency and type of intervention needed."*

- **Patient Interaction**

According to the question, Patient System generates an answer: *"The baby is 4 days old. The most recent total bilirubin level was 13.4 mg/dL, and the most recent direct bilirubin level was 0.5 mg/dL."*

- **Update Visit Log**

The selected question and answer pair are added to the visit log.

- **Dynamic Adjustment**

Central agent removes "Pediatric Hepatologist", current specialist list: "Pediatrician"

"Rationale": *"The newborn has an elevated total bilirubin but a low direct (conjugated) bilirubin, which suggests typical physiologic or possibly mild pathologic unconjugated hyperbilirubinemia. A pediatrician is well-equipped to manage and monitor neonatal jaundice at this stage. A pediatric hepatologist is not needed unless there is evidence of cholestasis or liver dysfunction (elevated direct bilirubin), which is not present here."*

- **Specialist Response Generation – Single agent**

Specialist: "Pediatrician"

Response Type: "Diagnosis",

Response Content: *"Physiologic jaundice of the newborn", "Breastfeeding jaundice",*

Rationale: *"A 4-day-old newborn with a total bilirubin of 13.4 mg/dL and a low direct (conjugated) bilirubin is most likely experiencing unconjugated hyperbilirubinemia. The most common causes at this age are physiologic jaundice, breastfeeding or breast milk jaundice, and hemolytic processes such as ABO or Rh incompatibility or hereditary red cell disorders. Other considerations include increased bilirubin production from bruising or cephalohematoma, polycythemia, and less commonly, infection. The low direct bilirubin makes cholestatic or obstructive causes less likely."*

- **Interaction Ends**

E Dataset Availability

Due to privacy constraints around MIMIC-derived data, we are unable to release the dataset in the supplementary materials. However, the MIMIC-Patient dataset is under review for release on PhysioNet and will be made publicly available upon acceptance.

F Potential Risk

While DynamiCare is designed for research and educational purposes, there are potential risks associated with the misuse or misinterpretation of its outputs. The system simulates clinical reasoning using language models that, despite strong performance, may generate incorrect, incomplete, or misleading responses if taken at face value. If such models were applied in real-world medical settings without appropriate oversight, they could lead to harmful diagnostic decisions or delayed treatments. Additionally, since MIMIC-Patient is derived from real patient data (via the de-identified MIMIC-III database), there is a need to ensure responsible data handling and prevent re-identification risks, even though direct identifiers have been removed. Finally, overreliance on open-ended AI-generated diagnoses could inadvertently reinforce biases or amplify gaps in model training data. We emphasize that our framework is not intended for clinical deployment and should be used strictly in controlled, academic environments.